

5-2 Milestone Four: Enhancement Three: Databases

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In this artifact milestone, I added Android's Room persistence library and implemented the necessary classes and annotations for two Data Access Objects (DAOs), one for PlotEntity and one for CropEntity. I added the necessary boilerplate to my Application class and AppContainer to create the Room database from a seed SQLite3 shipped in the app's assets directory, and hand-inserted several sample records.

I developed and implemented a database schema for Crops and Plots. These include establishing primary keys for each table, adding constraints for which fields must not be null, adding indexes for several fields I expect to be accessed frequently, and established a foreign key constraint (and deletion propagation policy) for a one-to-many relationship between PlotEntity and CropEntity. I was careful to maintain the schema so far in fifth normal form, by separating independent facts into different tables, for example separating Crops from their Watering and FertilizingRecords. I also began implementing much more thorough unit testing, specifically on the CropDao class.

While I feel that I have demonstrated the outcomes I set out to for this artifact, there is still much more to improve for my final project. For example, I have not yet converted the Watering, Fertilizing, Harvesting, or JournalRecord repositories to sqlite3-backed Room DAO yet. I have planned out the schema, but have not yet implemented the DAO interfaces and annotations yet in the Kotlin code itself.

Doing this artifact enhancement has been more enjoyable and faster than the previous ones. I have begun to grow accustomed with reading and writing Kotlin code and its various eccentricities and syntactic differences. I have wrapped my mind around the Android Jetpack

Compose architecture, and feel that I am starting to get the hang of reactive-based data flow and UI architecture. Unfortunately, I am still finding the official Android documentation for Compose, Compose Navigation, and Room to be confusing, out of date, and generally less than fully helpful. The best help I have found comes from copying parts of the documentation into an LLM and then challenging it with my questions, then trying to explain the concepts back to it in my own words while asking for its critiques.

AI Assistance Used

As discussed in the previous paragraph, I used LLMs to ask questions about code and documentation to help enhance my understanding and root out confusion. I used primarily gemma4:26b and gpt-oss:20b on a local server for this. I did not directly incorporate any code samples from any GenAI output into my application code.

References